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SIMILARITY OF NEURAL NETWORK MODELS: A SURVEY OF FUNCTIONAL AND REPRESENTATIONAL MEASURES

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ABSTRACT

Measuring similarity of neural networks to understand and improve their behavior has become an issue of great importance and research interest. In this survey, we provide a comprehensive overview of two complementary perspectives of measuring neural network similarity: (i) representational similarity, which considers how *activations* of intermediate layers differ, and (ii) functional similarity, which considers how models differ in their *outputs*. In addition to providing detailed descriptions of existing measures, we summarize and discuss results on the properties of and relationships between these measures, and point to open research problems. We hope our work lays a foundation for more systematic research on the properties and applicability of similarity measures for neural network models.

1 Introduction

Measures to quantify similarity of neural network models have been widely applied in the literature, usually to understand and improve deep learning systems. Examples include research on learning dynamics [101, 98], effects of width and depth [107], differences between supervised and unsupervised models [52], robustness [64, 104], effects of data and model updates [39, 86, 69, 99], evaluating knowledge distillation [134], designing ensembles [163], language representation [75, 53, 54], and generalizability [96, 79, 110].

However, understanding and measuring similarity of neural networks is a complex problem, as there are multiple perspectives on how such models can be similar. In this work, we specifically focus on two key perspectives: *representational* and *functional measures of similarity* (see Figure 1). Representational similarity measures assess how activations of intermediate layers differ, whereas functional similarity measures compare the outputs of neural networks with respect to their task. Both perspectives only provide a partial view on neural network similarity. Seemingly similar representations can still yield different outputs, and conversely, similar outputs can result from different representations. In that sense, combining these two complementary perspectives provides a more comprehensive approach to analyze similarity between neural networks at all layers.

Given the broad range of research on neural network similarity, numerous representational and functional similarity measures have been proposed and applied, often with lines of research being disconnected from each other. With this work, we provide a comprehensive overview of these two groups of similarity measures that gives a unified perspective on the existing literature and can inform and guide both researchers and practitioners interested in understanding and comparing neural network models.

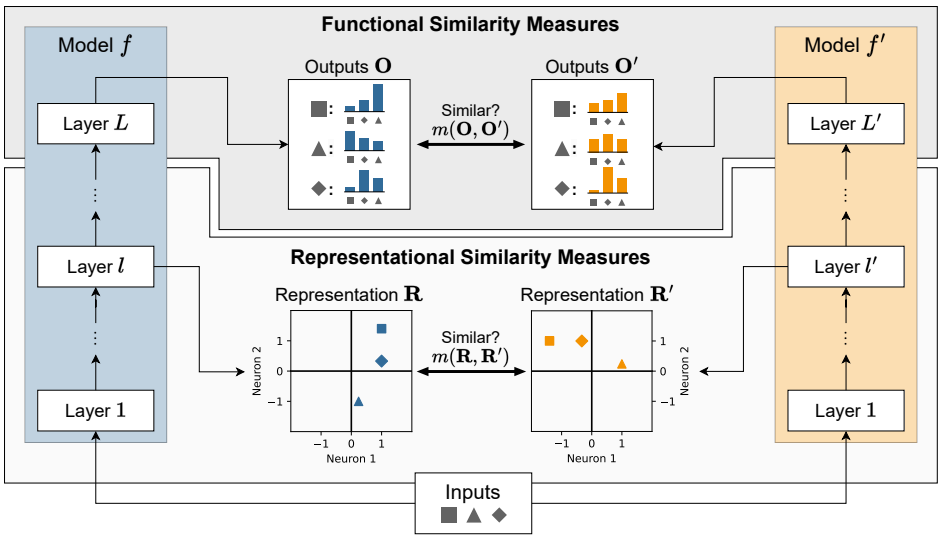

Figure 1: A conceptual overview of representational and functional similarity. We compare a pair of neural network models f, f'. Functional similarity measures mainly consider the outputs $\boldsymbol{O}, \boldsymbol{O'}$ of the compared models, whereas representational similarity measures consider their intermediate representations $\boldsymbol{R}, \boldsymbol{R'}$. All models get the same inputs. Specifically in classification tasks, outputs have clear and universal semantics, so that they can be compared in a straightforward manner. In contrast, the geometry of the representations requires more care when measuring their similarity. In the illustration above, for instance, rotating $\boldsymbol{R}$ by 90 degrees would yield an alignment of representations after which they would appear much more similar. Combined, representational and functional measures cover all layers of the models.

Measures for representational or functional similarity have been covered in prior work to some extent. Regarding representational similarity, measures for matrix correlation have been reviewed by [117, 160]. Existing surveys, however, lack coverage of more recent measures or do not consider the context of deep learning. A recent survey by Räuker et al. [118] reviews methods to interpret inner workings of neural networks, but discusses representational similarity measures only briefly. Sucholutsky et al. [136] complement this survey by discussing representational similarity with a focus on bringing together the communities of machine learning, neuroscience, and cognitive science, which have all been working independently on comparing representations. Functional similarity measures have been surveyed in the context of ensemble learning [76, 19], inter-rater agreement [6, 46, 143], model fingerprinting [138], and image and text generation scenarios [18, 22], which each focus on application scenarios with objectives different to our survey. We specifically focus on multi-class classification contexts for functional similarity measures.

To the best of our knowledge, our survey represents the first comprehensive review of representational and functional similarity measures for neural network models. This survey makes the following contributions:

1. **Systematic and comprehensive overview:** We formally define the problem of measuring representational and functional similarity in neural networks—the latter in the context of classification—and provide a systematic and comprehensive overview of existing measures.
2. **Unified terminology:** We provide detailed definitions, explanations, and categorizations for each measure in a unified manner, facilitating the understanding of commonalities and differences between measures.
3. **Analysis of practical properties and applicability:** We discuss the practical properties of existing measures, such as robustness to noise or confounding issues, and connections between existing measures to guide researchers and practitioners in applying these measures.
4. **Open research challenges:** We highlight unresolved issues of similarity measures and point out research gaps that can be addressed in the future to improve our understanding of neural networks in general.

While we focus on measures for representational and functional similarity due to their prevalence and general applicability, we acknowledge various other approaches to comparing neural networks. In particular, the measures covered in our survey differ from methods typically used to assess and optimize similarity during model training. We discuss these and other approaches in Appendix E.

2 Similarity of Neural Network Models

We consider the problem of comparing neural networks, which we assume to have the form

$$f = f^{(L)} \circ f^{(L-1)} \circ \cdots \circ f^{(1)}, \tag{1}$$

with each function $f^{(l)} : \mathbb{R}^{D^{(l-1)}} \longrightarrow \mathbb{R}^{D^{(l)}}$ denoting a single layer of $D := D^{(l)}$ neurons, and a total number of $L \in \mathbb{N}$ layers. These networks operate on a set of N given inputs $\{\boldsymbol{X}_i\}_{i=1}^N$, which we typically assume to be vectors in $\mathbb{R}^p$, $p \in \mathbb{N}$, although these can also be higher-dimensional structures as occurring in image or video data. We collect these inputs in a matrix $\boldsymbol{X} \in \mathbb{R}^{N \times p}$ so that the i-th row $\boldsymbol{X}_i$ corresponds to the i-th input. To further simplify notation, we also denote individual inputs $\boldsymbol{X}_i$ as *instances* $i \in \{1, \ldots, N\}$. We generally do not make any assumption about the number of features p, the depth of the network L, the width or activation function of any layer $f^{(l)}$, or the training objective.

Similarity of neural network models is then quantified by *similarity measures* m. For simplicity, we also consider measures that quantify *distance* between models as similarity measures, since these concepts are generally equivalent. In our survey, we specifically consider two kinds of similarity, namely *representational similarity* and *functional similarity*. Representational similarity measures consider how the inner activations of neural network models differ, whereas functional similarity measures compare the output behavior of neural networks with respect to a given (classification) task. Combined, these two notions allow for nuanced insights into similarity of neural network models [137, 69, 52].

In the following, we give more thorough definitions of representational and functional similarity. For the rest of this paper, we introduce notations for commonly used variables only once. In Appendix A, we provide an overview of notation and several definitions of variables and functions that are used in this paper.

2.1 Representational Similarity

Representational similarity measures compare neural networks by measuring similarity between activations of a fixed set of inputs at any pair of layers. Given such inputs $\boldsymbol{X}$, we define the representation of model f at layer l as a matrix

$$\boldsymbol{R} := \boldsymbol{R}^{(l)} = \left(f^{(l)} \circ f^{(l-1)} \circ \cdots \circ f^{(1)}\right)(\boldsymbol{X}) \in \mathbb{R}^{N \times D}. \tag{2}$$

The activations of instance i then correspond to the i-th row $\boldsymbol{R}_i = \left(f^{(l)} \circ \cdots \circ f^{(1)}\right)(\boldsymbol{X}_i) \in \mathbb{R}^D$, which we denote as *instance representation*. The activations of single neurons over all instances correspond to the columns of $\boldsymbol{R}$, and we denote the j-th column of $\boldsymbol{R}$ as $\boldsymbol{R}_{-,j}$. Like the inputs, we also consider the instance representations $\boldsymbol{R}_i$ to be vectors even though in practice, e.g., in convolutional neural networks, these activations can also be matrices. In such a case, these representations can be flattened (see Appendix B).

Representational similarity measures are typically defined as mappings $m : \mathbb{R}^{N \times D} \times \mathbb{R}^{N \times D'} \longrightarrow \mathbb{R}$ that assign a similarity score $m(\boldsymbol{R}, \boldsymbol{R'})$ to a pair of representations $\boldsymbol{R}, \boldsymbol{R'}$, which are derived from different models f, f', but use the same inputs $\boldsymbol{X}$. While we assume here that representations stem from different models, representational similarity measures can also be used to compare representations of different layers of the same model. Without loss of generality, we assume that $D \leq D'$, though some measures require that $D = D'$. In such cases, preprocessing techniques can be applied (see Appendix B). We note that this definition is limited to comparisons of *pairs* of representations, which is the standard setting in literature. In practice, one may also be interested in measuring similarity of *groups* of representations. The most direct way to obtain such measures of similarity for groups of representations from the standard pairwise measures is to aggregate the pairwise similarity scores, e.g., by averaging the similarities of all pairs of representations.

Typical issues when measuring similarity of representations are that the measures have to identify when a pair of representations is equivalent, and that some measures may require preprocessing of the representations. In the following sections, we discuss these issues and related concepts in more detail.

Equivalence of Representations. Even if two representation matrices $\boldsymbol{R}, \boldsymbol{R'} \in \mathbb{R}^{N \times D}$ are not identical on an element-per-element basis, one may still consider them to be equivalent, i.e., perfectly similar. An intuitive example for such a case would be when representations only differ in their sign, i.e., $\boldsymbol{R} = -\boldsymbol{R'}$, or when representations can be rotated onto another. Such notions of equivalence can be formalized in terms of bijective mappings (transformations) $\varphi : \mathbb{R}^{N \times D} \longrightarrow \mathbb{R}^{N \times D}$ that yield $\varphi(\boldsymbol{R}) = \boldsymbol{R'}$. What kind of transformations constitute equivalence between representations may vary depending on the context at hand. For instance, equivalence up to rotation does not make sense if some feature dimensions are already aligned with fixed axes, as is the case in interpretable word embeddings where axes may represent scales between polar opposites like “bright" and “dark" [94]. Thus, we define equivalence of representations in terms of groups of transformations $\mathcal{T} := \mathcal{T}(N, D)$, and call two representations $\boldsymbol{R}, \boldsymbol{R'}$ equivalent with respect to a group $\mathcal{T}$, written as $\boldsymbol{R} \sim_{\mathcal{T}} \boldsymbol{R'}$, if there is a $\varphi \in \mathcal{T}$ such that $\varphi(\boldsymbol{R}) = \boldsymbol{R'}$.